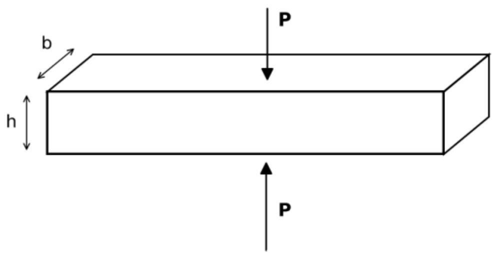
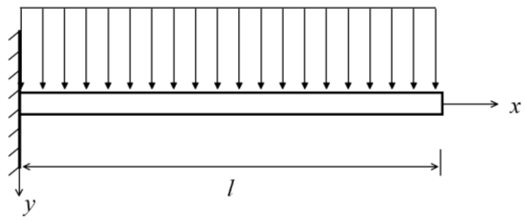
Variation Method & FEA Course
Homework 3

Submitted before June 15, 2026

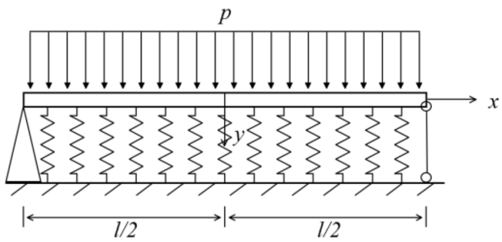
1. As shown in the picture below, applying a pair of holding force **P** at the middle of a rod with the rectangle section. Calculate the length change Δ of the rod.



2. Prove that $w = a(1 - cos\frac{\pi x}{2l})$ is not the trial function of the Galerkin method for the cantilever beam below.



3. Obtain the differential equations and boundary conditions of this elastic foundation beam by the variation method. Solve it using first order approximation of Galerkin method.



4. Establish the shape functions of a two-node Euler-Bernuolli beam element.